

COMP 356 Practice Final Exam

name:

1. (10 points) What output does the following C++ program produce?

```
#include <iostream>

class foo {
public:
    foo() {}
    virtual void print1() {std::cout << "class foo" << std::endl;}
    void print2() {std::cout << "class foo" << std::endl;}
};

class bar: public foo {
public:
    bar() {}
    virtual void print1() {std::cout << "class bar" << std::endl;}
    void print2() {std::cout << "class bar" << std::endl;}
};

int main() {
    foo *fooptr;
    foo f;
    bar b;

    fooptr = new foo();
    fooptr->print1();
    fooptr = new bar();
    fooptr->print1();
    fooptr->print2();
    f = b;
    f.print1();
    f.print2();
}
```

2. (10 points) Write a Prolog relation **subset** that determines whether its first argument (a list representing a set) is a subset of its second argument (also a list representing a set). For example, the queries:

```
subset([], [3, 4]).
subset([4, 3, 4], [4, 3, 2, 1]).
```

should both succeed, and the query:

```
subset([4, 3, 1], [4, 3, 2]).
```

should fail.

3. (15 points) Write a Scheme function `merge` that takes two sorted lists of numbers as arguments and returns the result of merging them into one list. For example, the call: `(merge '(1 4 7 9) '(3 4 5))` should return: `(1 3 4 4 5 7 9)`.

4. (12 points) Write a Scheme function `countP` that takes a list and a function as arguments. The function argument should take one argument and return boolean. The result of a call to `countP` should be the number of elements of the list that make the function argument true. For example:

```
(countP '(1 4 3 0 6 5) (lambda (x) (< x 3)))
```

should return 2.

5. Consider the following declarations.

```
typedef struct foo {  
    int x;  
    double y;} footype;  
typedef struct bar {  
    int x;  
    double y;} bartype;  
  
typedef footype atype[10];  
typedef bartype btype[10];  
typedef atype ctype; typedef btype dtype; typedef ctype etype;  
  
atype a1; btype a2; ctype a3; dtype a4; etype a5; footype a6; bartype a7;
```

(a) (6 points) List all variables whose type is structurally compatible with the type of **a1**.

(b) (6 points) List all variables whose type is declaration compatible with the type of **a1**.

6. (8 points) Write an EBNF definition that describes the set of all strings 0's and 1's that contain at most three 1's.

7. (7 points) Translate the following Prolog rule into a statement of first order logic.

```
sibling(X, Y) :- parent(Z, X), parent(Z, Y).
```

8. (8 points) Write a tail recursive version of the following Scheme function. Include an “interface” function that calls your tail recursive function with the appropriate initial value for the accumulator parameter.

```
(define sumlist (lambda (lst)
  (if (null? lst) 0
      (+ (car lst) (sumlist (cdr lst))))))
```

9. (8 points) Identify at least two errors in the following Java class definition.

```
import java.io.*;

public abstract class MyClass {
  private int i;
  private int j;

  public MyClass() {}

  public MyClass(int i, int j) {
    this.i = i;
    this.j = j;
  }

  public abstract void print() {
    System.out.println(i);
    System.out.println(j);
  }

  public static void main(String argv[]) {
    MyClass m = new MyClass(2, 3);
    m.print();
  }
}
```

10. (6 points) Given the following BNF definition:

```
<typeddecl> → <type> <idlist> ;  
<type> → int | float | bool  
<idlist> → <id> | <id>, <idlist>  
<id> → x | y | z
```

Draw the parse tree for the string: int x, y;

11. (12 points) Why does multiple inheritance of interfaces cause fewer problems than multiple inheritance of classes? Give at least two reasons, and justify each reason.